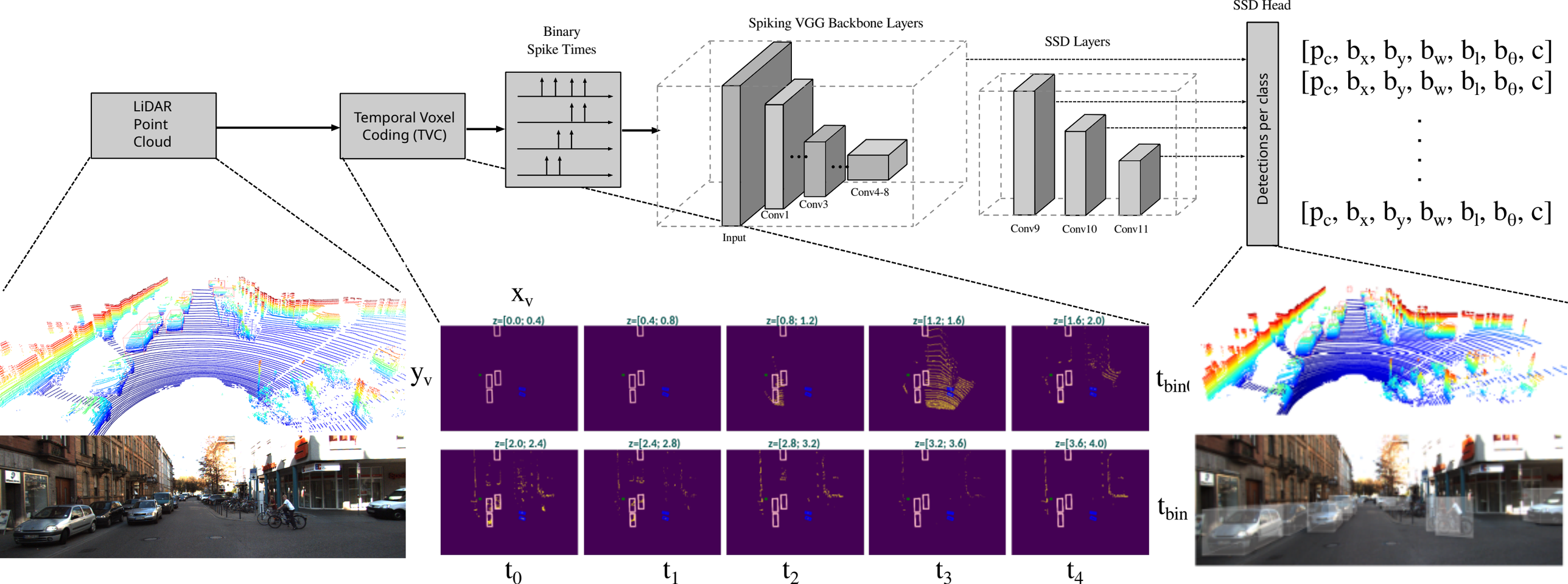

LiDAR Point Cloud
Temporal Voxel Coding (TVC)
Binary Spike Times
Spiking VGG Backbone Layers
Input
Conv1
Conv3
Conv4-8
SSD Layers
Conv9
Conv10
Conv11
SSD Head
Detections per class
[p_c, b_x, b_y, b_w, b_l, b_θ, c]
[p_c, b_x, b_y, b_w, b_l, b_θ, c]
[p_c, b_x, b_y, b_w, b_l, b_θ, c]
x_v
y_v
z=[0.0; 0.4)
z=[0.4; 0.8)
z=[0.8; 1.2)
z=[1.2; 1.6)
z=[1.6; 2.0)
z=[2.0; 2.4)
z=[2.4; 2.8)
z=[2.8; 3.2)
z=[3.2; 3.6)
z=[3.6; 4.0)
t_0
t_1
t_2
t_3
t_4
t_bin
t_bin